



TOP 23

WIRESHARK FILTERS



for Threat Detection



Rajneesh Gupta
@rajneeshcyber

WHAT IS WIRESHARK?

Wireshark is a widely-used open-source network protocol analyzer.

Purpose

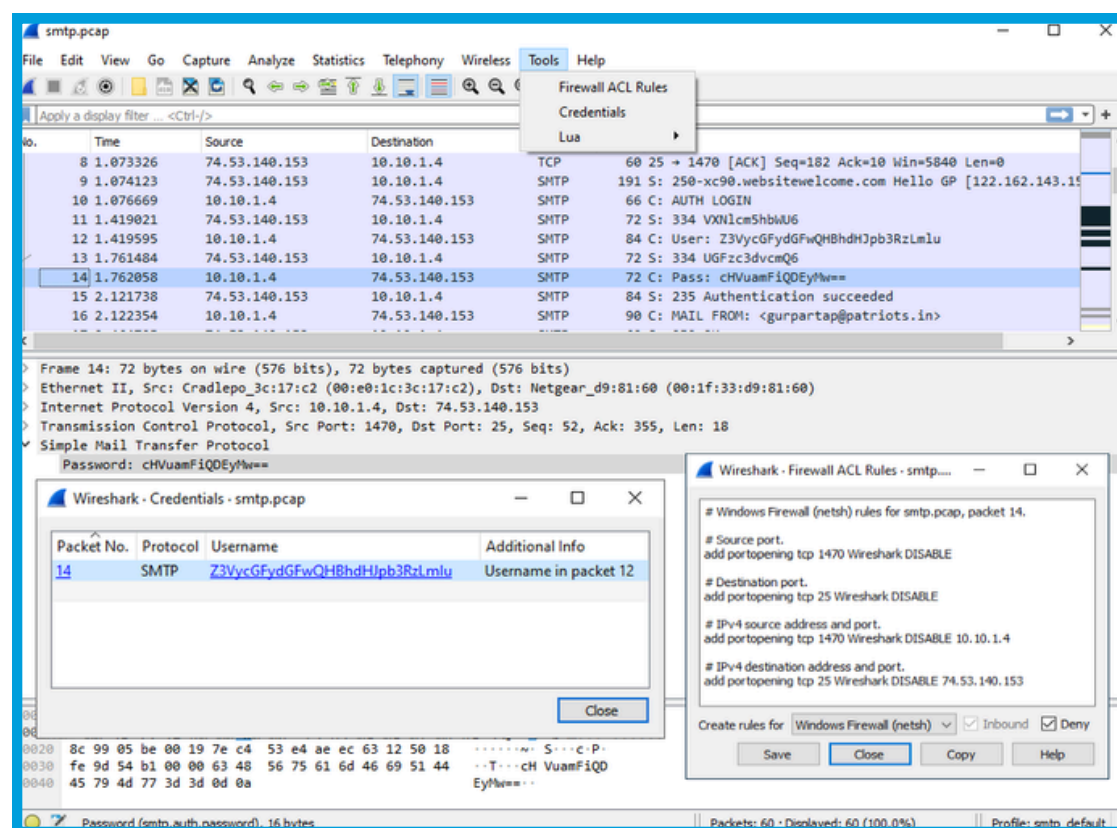
It captures and analyzes network traffic in real-time, helping in troubleshooting, security analysis, and network optimization.

Key Features:

- Deep inspection of hundreds of protocols.
- Live traffic capture and offline analysis.
- Ability to decrypt many protocols, including SSL/TLS.

WHY USE WIRESHARK?

- **Network Troubleshooting:** Identify network issues, bottlenecks, and performance problems.
- **Security Analysis:** Detect malicious activities, such as malware, DDoS attacks, and suspicious traffic.
- **Protocol Development:** Assist developers in debugging protocol implementations.
- **Real-Time Monitoring:** Monitor network data in real-time for quick response to anomalies.



WIRESHARK FILTERS

What are Filters?

Filters in Wireshark help narrow down specific traffic of interest from a large dataset.

Types of Filters:

- Capture Filters: Applied during data capture, limiting the data being captured.
- Display Filters: Applied after capturing data, allowing for in-depth analysis of specific traffic.

Common Display Filter Examples:

- IP Filter: `ip.addr == 192.168.1.1` (Shows traffic to/from a specific IP address)
- Port Filter: `tcp.port == 80` (Shows traffic on a specific port, like HTTP)
- Protocol Filter: `http` (Displays HTTP protocol traffic)

TOP 23 WIRESHARK FILTERS

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HTTP GET FLOODING

How does the threat work?

Attackers generate a high volume of HTTP GET requests, overloading the server's capacity, causing denial of service.

Display filter

```
http.request.method == "GET"
```

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Check the HTTP headers for an abnormal number of requests from the same source IP.
- If the source IP varies, investigate suspicious patterns in user-agent strings.

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DNS TUNNELING

How does the threat work?

DNS queries are manipulated to transmit data, bypassing firewalls and performing covert data exfiltration.

Display filter



How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Inspect the DNS query field for abnormal payload sizes or repeated requests.
- For high-frequency DNS queries from one source, analyze the DNS response time for unusual delays.

SMB BRUTE FORCE

How does the threat work?

Multiple login attempts are made via the SMB protocol, trying to guess valid credentials through repetitive trials.

Display filter

A dark gray rectangular box representing a Wireshark display filter. It has three colored circles (red, orange, green) in the top right corner. The text 'smb.cmd == 0x73' is written in white.

```
smb.cmd == 0x73
```

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Check the SMB headers for failed login attempts in the status code field.
- If the attack includes successful logins, monitor SMB command response times.

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SYN FLOOD ATTACK

How does the threat work?

Attackers flood the target server with SYN packets without completing the TCP handshake, exhausting server resources.

Display filter

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Review the TCP headers and check if there's a high volume of SYN packets without corresponding ACKs.
- If there are some ACK responses, look for unusually delayed ACKs or connection resets.

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DNS AMPLIFICATION

How does the threat work?

Attackers use a small request to trigger large DNS responses, reflecting traffic back to the target and overwhelming it.

Display filter

A dark gray rectangular box representing a Wireshark display filter bar. It has three colored circles (red, orange, green) in the top right corner. The text 'dns.qry.name' is written in white inside the box.

dns.qry.name

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Check the DNS response section for large-sized responses and unexpected source addresses.
- If response sizes seem normal, look at the TTL field for unusually low values.

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ICMP FLOODING (PING OF DEATH)

How does the threat work?

The attacker sends numerous ICMP requests (pings) to overload a device, consuming bandwidth and resources.

Display filter

A dark gray rectangular window with rounded corners, representing a Wireshark display filter. It has three small colored circles (red, orange, green) in the top right corner. The text 'icmp' is displayed in white on the left side.

icmp

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Review the ICMP headers for a high frequency of Echo Request packets from a single source.
- If packet sizes are unusual, inspect for ICMP packets larger than the standard 64 bytes.

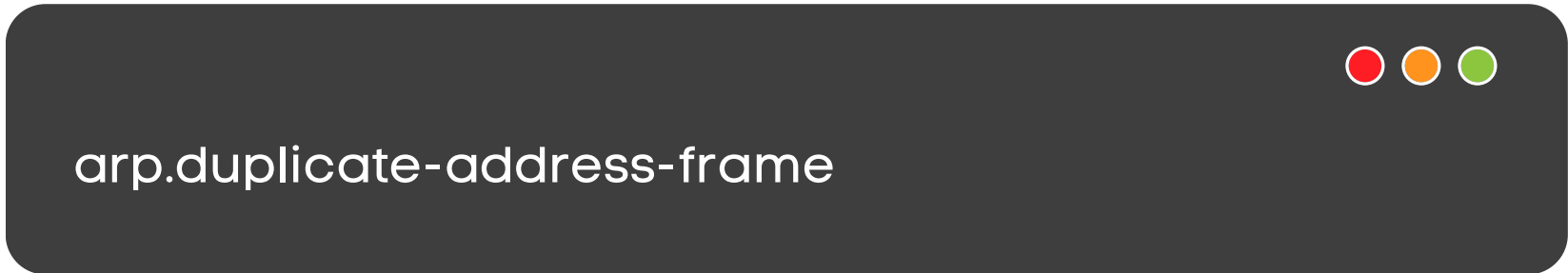


ARP SPOOFING

How does the threat work?

An attacker sends falsified ARP messages, leading devices to link wrong MAC addresses with IP addresses.

Display filter

A dark gray rectangular window with rounded corners, representing a Wireshark display filter. In the top right corner, there are three small circles: red, orange, and green. The text 'arp.duplicate-address-frame' is centered in the window.

```
arp.duplicate-address-frame
```

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Inspect the ARP header for mismatches between IP and MAC addresses.
- If there are no clear mismatches, check for ARP requests with identical source IPs but different MAC addresses.

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DNS POISONING

How does the threat work?

DNS responses are altered or forged to redirect users to malicious websites instead of legitimate ones.

Display filter

```
dns.flags.rcode != 0
```

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Analyze DNS response headers for mismatched IP addresses or altered TTL values.
- If there's no direct mismatch, check the response time field for unexpected delays.

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SUSPICIOUS HTTP USER-AGENT

How does the threat work?

Attackers may use irregular User-Agent strings to mask their identity or avoid detection during attacks.

Display filter

A dark gray rectangular box representing a Wireshark display filter bar. It has three colored circles (red, orange, green) in the top right corner. The text 'http.user_agent' is written in white inside the box.

```
http.user_agent
```

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Examine the HTTP headers for unusual or malformed User-Agent strings.
- If the User-Agent is not obviously suspicious, correlate requests with known malicious IP addresses.

FTP PLAINTEXT AUTHENTICATION

How does the threat work?

FTP transmits login credentials in plaintext, making them vulnerable to interception by attackers.

Display filter

```
ftp.request.command == "USER"
```

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Review FTP request headers for visible usernames and passwords.
- If no clear credentials are visible, inspect packet payloads for plaintext data in the FTP stream.

BRUTE FORCE LOGIN (SSH)

How does the threat work?

Attackers repeatedly attempt to guess valid SSH credentials, exploiting weak or default passwords.

Display filter

A dark gray rectangular box representing a terminal window. In the top right corner, there are three small circles: red, orange, and green. On the left side of the box, the text 'ssh' is displayed in a white, monospaced font.

ssh

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Look for repetitive login attempts in the SSH connection handshake fields.
- If there are fewer attempts, analyze the session duration for failed vs successful logins.

DHCP STARVATION ATTACK

How does the threat work?

Attackers request multiple DHCP leases, depleting available IP addresses to prevent legitimate network access.

Display filter

A dark gray rectangular bar representing a Wireshark display filter. It has three colored circles (red, orange, green) in the top right corner. The text 'dhcp' is written in white on the left side.

dhcp

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Analyze the DHCP transaction ID for a high number of requests without corresponding DHCP Offers.
- If offers are present, check for quick lease time expiration in the DHCP header.

SUSPICIOUS TLS CERTIFICATE

How does the threat work?

Attackers use invalid or self-signed TLS certificates to compromise encrypted communications.

Display filter

```
ssl.handshake.type == 11
```

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Review the SSL handshake header for certificates signed by unknown authorities.
- If the certificate seems valid, verify the encryption protocol used for downgrade attempts.

TELNET CLEARTEXT TRANSMISSION

How does the threat work?

Telnet transmits all data, including passwords, in plaintext, making it easily interceptable by attackers.

Display filter

A dark gray rectangular box representing a Wireshark display filter bar. It has three colored circles (red, orange, green) in the top right corner. The text 'telnet' is written in white on the left side.

telnet

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Check for visible login credentials in the Telnet data fields.
- If no credentials are visible, inspect for suspicious command execution strings in the payload.

UNAUTHORIZED RDP ACCESS

How does the threat work?

Attackers attempt unauthorized access to a system via Remote Desktop Protocol (RDP) to take control of the system.

Display filter

A dark gray rectangular box representing a Wireshark display filter. It has three colored circles (red, orange, green) in the top right corner. The text 'tcp.port == 3389' is written in white inside the box.

```
tcp.port == 3389
```

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Inspect RDP header fields for repeated connection attempts from the same IP.
- If the source IP changes, review session IDs for abnormal session initiation patterns.

PORT SCANNING

How does the threat work?

Attackers scan for open ports on a system to find vulnerabilities and identify services to exploit.

Display filter

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Review the TCP flags in headers for SYN packets across different ports.
- If SYN-ACK responses are present, inspect for suspicious response delays or resets.

SMTP EXFILTRATION

How does the threat work?

Sensitive information is extracted from a network by sending it out via email using the SMTP protocol.

Display filter

A dark gray rectangular box representing a Wireshark display filter bar. In the top right corner, there are three small colored circles: red, orange, and green. The text 'smtp' is written in white on the left side of the bar.

smtp

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Review SMTP headers for large or abnormal email attachments.
- If attachment size seems normal, inspect destination domains for unauthorized or unknown email addresses.

ROGUE DHCP SERVER

How does the threat work?

A rogue DHCP server assigns unauthorized IP addresses, redirecting network traffic or causing denial of service.

Display filter

A dark gray rectangular window representing a Wireshark display filter. It has three colored circles (red, yellow, green) in the top right corner. The text 'dhcp' is displayed in white on the left side.

dhcp

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Review the DHCP Offer packets and compare the DHCP server IP address to authorized DHCP servers.
- If server IPs match, check for abnormal lease durations or lease renewals in the DHCP headers.

SQL INJECTION ATTACK

How does the threat work?

Malicious SQL code is injected into vulnerable input fields, granting attackers unauthorized access to the database.

Display filter

A dark gray rectangular box with rounded corners, representing a Wireshark display filter. In the top right corner, there are three small circles in red, orange, and green. The text "http.request.uri contains \"SELECT\"" is centered in white.

```
http.request.uri contains "SELECT"
```

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Review HTTP request URIs for SQL commands like SELECT, DROP, or INSERT within the URL or payload.
- If no SQL commands are found, inspect HTTP headers for error responses that could indicate failed injection attempts.

VOIP EAVESDROPPING (SIP)

How does the threat work?

Attackers intercept Voice over IP (VoIP) traffic, allowing them to listen in on private conversations.

Display filter

A dark gray rectangular box representing a Wireshark display filter bar. It has three colored circles (red, yellow, green) in the top right corner. The text 'sip' is written in white on the left side.

sip

How to detect?

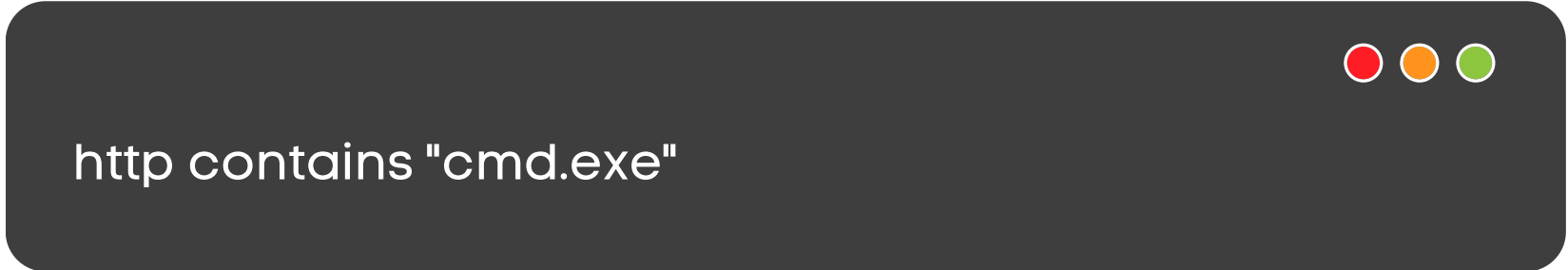
- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Check the SIP headers for unauthorized call setups or unknown user IDs initiating the call.
- If calls seem normal, analyze the RTP streams for abnormal traffic patterns or packet loss

MALWARE COMMAND & CONTROL TRAFFIC

How does the threat work?

Malware communicates with a Command & Control (C2) server to receive instructions for further malicious activity.

Display filter

A dark gray rectangular box with rounded corners, representing a Wireshark display filter. In the top right corner, there are three small circles: red, orange, and green. The text "http contains \"cmd.exe\"" is displayed in white.

http contains "cmd.exe"

How to detect?

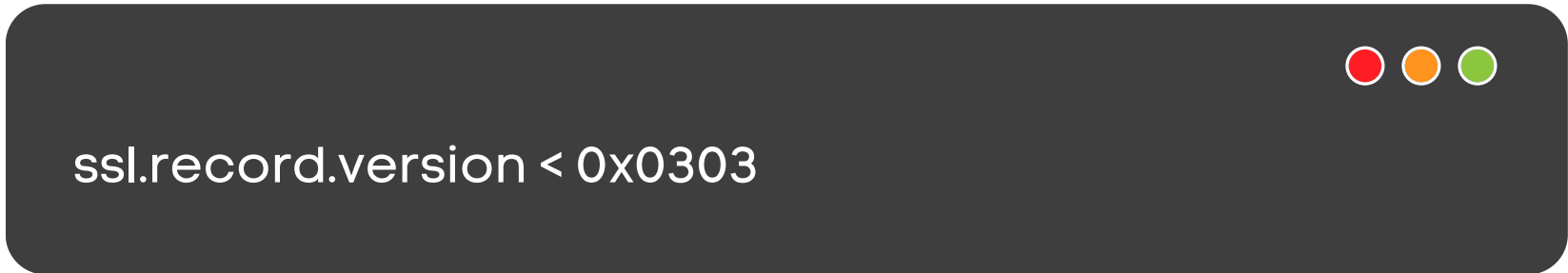
- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Review the HTTP request payloads for command execution instructions such as cmd.exe.
- If no commands are found, check for suspicious external server connections, especially to uncommon IPs.

SSL DOWNGRADE ATTACK

How does the threat work?

Attackers force the use of weaker SSL/TLS encryption protocols, reducing security and allowing for interception or decryption.

Display filter

A dark gray rectangular box with rounded corners, representing a Wireshark display filter. In the top right corner, there are three small circles: red, orange, and green. The text 'ssl.record.version < 0x0303' is written in white monospace font in the center of the box.

```
ssl.record.version < 0x0303
```

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Check the SSL/TLS handshake headers for negotiation with older protocol versions like SSL 2.0 or 3.0.
- If SSL/TLS versions are acceptable, inspect certificate chains for self-signed or expired certificates.

SUSPICIOUS HTTP POST REQUESTS

How does the threat work?

Attackers use HTTP POST requests to upload files or send data to a server, which may be used for data exfiltration or malicious uploads.

Display filter

```
http.request.method == "POST"
```

How to detect?

- Open Wireshark, navigate to the filter bar, and enter the above-mentioned display filter.
- Review HTTP headers for POST requests with unusually large payloads or requests to unknown IP addresses.
- If payload sizes are small, inspect the content of the POST body for base64-encoded or binary data.

CONCLUSION

In conclusion, using Wireshark with targeted filters allows for effective threat detection and network security monitoring.

- Flexible tool: Wireshark adapts to various threats.
- Real-time detection: Spot suspicious traffic instantly.
- Layered insight: Analyze threats across multiple layers.
- Proactive defense: Identify attacks early to prevent damage.
- Customizable filters: Tailor filters to reduce false positives.
- Continuous updates: Stay prepared with regular Wireshark improvements.



Reach us at
hi@haxsecurity.com

Security Consulting

- Risk assessment
- Security Architecture
- SOC Set up

Penetration testing

- Internal Pentest
- External Pentest
- Web App Pentest

Training and Courses

- SOC Training
- Certification Training
- Vendor-specific learning

Labs

- Hands-on Labs
- Career Path Labs
- Cyberrange for businesses